Insights and Recommendations: Olist Delivery Performance

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Summary

Late delivery affects a small group of orders, not the whole business. 8.1% of delivered orders (7,826 of 96,470) arrive after the promised date, and those orders score 2.57 on average against 4.29 for on-time orders.

Orders arrive about 11 days before the promised date on average (12.6 days actual, 23.7 days promised). So the fix is to deal with the orders that run very late, not to make every promise longer.

Nearly three in four late orders (73%) are cross-state orders. The worst late rates are in AL (23.9%), MA (19.7%), PI (16.0%) and CE (15.3%). Most of the extra time is lost in transit, not in seller handling.

Key findings

Base for all figures: 96,470 delivered orders with a customer delivery date. Late means delivered after the date promised at purchase.

#	Finding	Evidence	Where it comes from
1	8.1% of delivered orders are late	7,826 of 96,470; 5,164 (5.35%) are more than 3 days late	Dashboard cards; SQL 02 Check A
2	Review score falls as lateness grows	On time 4.29 · 1-3 days 3.76 · 3-7 days 2.32 · 7-14 days 1.75 · over 14 days 1.71	Dashboard delay chart; SQL 02 Check B
3	Late orders are far more likely to get 1-2 stars	54.1% of reviewed late orders vs 9.2% of reviewed on-time orders	Excel Analysis sheet; SQL 03 Q3

#	Finding	Evidence	Where it comes from
4	Cross-state orders are late more often and take twice as long	9.27% late and 15.15 days vs 6.06% and 7.95 days for same-state; 73% of late orders (5,723) are cross-state	Dashboard route chart; SQL 02 Check C; SQL 04
5	Late rates differ strongly by destination state	AL 23.9% · MA 19.7% · PI 16.0% · CE 15.3% · SE 15.2%; SP is 5.9%. SP and RJ together hold 52% of all late orders because of their volume	SQL 03 Q1; SQL 04 (the dashboard chart shows the 10 busiest states only)
6	Lateness peaks in three months	Nov 2017 14.3% · Feb 2018 16.0% · Mar 2018 21.4%, against 8.13% for the whole Jan 2017 to Aug 2018 window	Dashboard monthly chart; SQL 03 Q2
7	A few sellers are very late	621 sellers have 30+ delivered orders (8.08% late combined); the five worst are 27% to 35% late	Excel seller pivot; SQL 03 Q4 and Q5

Where the time is lost

About 86% of the extra time on late orders is spent in transit, after the parcel is handed to the carrier. Seller handling adds only about 3 days.

Stage (average days)	On-time orders	Late orders	Difference
Purchase to carrier handover (seller side)	3.0	5.8	+2.8
Carrier handover to customer (transit)	7.9	25.7	+17.8
Orders handed over after the seller's shipping limit date	7.4% (6,595 of 88,644)	27.2% (2,128 of 7,825)	3.7 times as likely

Two points follow. Missing the shipping limit date is common among late orders, so seller handover still matters. But the transit leg is where most of the delay builds up.

This breakdown comes from SQL only (file 04) and is not on the Excel dashboard. One late order has no carrier date and is left out of this table.

Recommendations

Five actions, in priority order. The owners are assumed roles, because this is a public dataset with no real company behind it. I have not set targets, because that would need cost data and input from the business.

#	Action	Based on	Assumed owner	How to measure
1	Review carrier performance on the cross-state lanes into AL, MA, PI and CE first, then RJ because of its volume	Findings 4 and 5; transit is 86% of the extra time	Logistics and carrier management	Weekly late rate and transit days by destination state (baseline: AL 23.9%, MA 19.7%, PI 16.0%, CE 15.3%)
2	Set up an early-warning alert for orders likely to pass their promised date, and tell the customer before it happens	Finding 2: score drops 0.5 points when 1-3 days late, but about 2 points beyond 3 days	Customer service	Review score of late orders (baseline 2.57); share of orders more than 3 days late (baseline 5.35%)
3	Track seller handover against the shipping limit date and warn sellers who miss it often	Late orders miss it 3.7 times as often as on-time orders	Seller management	Share of orders handed over after the limit date (baseline 9.0% overall)
4	Start with a performance review of the worst sellers (five sellers are 27% to 35% late)	Finding 7	Seller management	Late rate per seller with 30+ orders

#	Action	Based on	Assumed owner	How to measure
5	Check capacity and promised dates before peak months	Finding 6: Nov 2017, Feb 2018 and Mar 2018	Operations planning	Monthly late rate against the 8.1% average

Recommendations 2 and 5 rest on associations in the data, so they are hypotheses to test, not proven fixes. The data cannot show why November, February and March are worse.

Risks and limits

This analysis shows what goes together in the data. It does not prove that late delivery causes low scores.

- **Association, not cause.** Product quality, price and the seller's service also affect reviews, and none of them is in the data.
- **Reviewed orders only.** Review figures cover 95,824 of the 96,470 delivered orders. The first review per order is used, which drops 551 extra review rows.
- **"Late" depends on the promised date.** The promise is 11 days longer than the actual time on average, so a stricter promise would change the late rate.
- **Route type uses the first item's seller.** Orders with items from several sellers are classed by the first one only.
- **Carriers are not named in the data.** Recommendation 1 can point to destination states and lanes, not to a specific carrier.
- **Old data from a single source.** The orders run from 2016 to 2018 and the dataset is a public sample, so current performance may differ.
- **No cost data.** The financial value of fixing late deliveries cannot be estimated.

Traceability

Every finding can be traced to the Excel workbook and to the SQL files. The SQL files were re-run against the database to confirm each number here.

Finding	Excel (dashboard view / sheet)	SQL file and query	Recommendation
1. Late rate 8.1%	Dashboard cards; KPI_Checks	02 Check A	2, 5
2. Score falls with delay	Chart "Orders and review score by delay"	02 Check B	2
3. 1-2 star share	Analysis sheet	03 Q3	2
4. Cross-state vs same-state	Chart "Same-state vs cross-state"	02 Check C; 04	1
5. States	Chart "10 busiest states"; Pivots	03 Q1; 04	1
6. Peak months	Chart "Late rate by month"	03 Q2	5
7. Sellers	Pivot_Sellers	03 Q4 and Q5	3, 4
Where the time is lost	Not in Excel	04	1, 3

Data: Olist Brazilian E-Commerce Public Dataset on Kaggle, licence CC BY-NC-SA 4.0, used for education only.

How this was prepared. This document was prepared with the assistance of Claude, an AI assistant from Anthropic. Claude made the documentation process smoother by suggesting the industry-standard structure, helping with the wording, and checking that the numbers match across the Excel, SQL and Power BI work. It also answered my questions and pointed out my mistakes. The data analysis in Excel, SQL and Power BI, the dashboard, the conclusions and the diagrams were done by me. Claude helped me with some of the SQL queries, which I ran and checked against my Excel results. Not on the dashboard: median and 90th-percentile delivery time, a product-category view and a reference line on the monthly chart. All three are listed as known gaps in the Requirements Specification.